Experiment 5: DBMS

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1. Aim of the Session

To demonstrate the creation of a database schema, population of records, and the use of the CASE Expression to perform conditional analysis on numerical data (parity check).

2. Objective of the Session

- To learn how to manage table lifecycles using DROP and CREATE.
- To understand the application of the modulo operator (%) in SQL.
- To gain proficiency in creating virtual columns using CASE-WHEN-THEN-ELSE logic.
- To practice basic Data Manipulation Language (DML) and Data Query Language (DQL) operations.

3. Practical / Experiment Steps

The experiment consists of the following logic:

1. Table Setup: Ensure a clean environment by removing any existing employee table and creating a new one with specific constraints (Primary Key).
2. Data Population: Insert six distinct records representing employee IDs, names, and salaries.
3. Conditional Querying: Execute a SELECT statement that calculates the remainder of the salary divided by 2 (salary%2).
4. Classification: If the remainder is 0, classify as 'Even Salary'; otherwise, classify as 'Odd Salary' using a CASE statement.

4. Procedure of the Practical

(i) Start your SQL environment (e.g., MySQL, PostgreSQL, or SQL Server Management Studio).

(ii) Open a new Query Editor window.

(iii) Create or select the required database schema.

(iv) Type the DROP and CREATE commands to define the table structure.

(v) Execute the INSERT commands to populate the table with the provided dataset.

(vi) Write the SELECT query incorporating the CASE expression for salary classification.

(vii) Execute the script and verify that the salary_type column correctly reflects the parity of the salary.

(viii) Note down the results and take a screenshot of the output grid.

5. I/O Analysis (Input / Output Analysis)

Input: Employee Table Data

emp_id	emp_name	salary
101	alice	50024
102	akash	25001
103	bimlesh	45120
104	charlie	48203
105	david	68103
106	kelly	79536

Output: Query Results

emp_id	emp_name	salary	salary_type
101	alice	50024	Even Salary
102	akash	25001	Odd Salary
103	bimlesh	45120	Even Salary
104	charlie	48203	Odd Salary
105	david	68103	Odd Salary

emp_id	emp_name	salary	salary_type
106	kelly	79536	Even Salary

6. **Output Screenshot:**

	emp_id [PK] integer	emp_name character varying (50)	salary integer	salary_type text
1	101	alice	50024	Even Salary
2	102	akash	25001	Odd Salary
3	103	bimlesh	45120	Even Salary
4	104	charlie	48203	Odd Salary
5	105	david	68103	Odd Salary
6	106	kelly	79536	Even Salary

7. Learning Outcome

Through this practical session, I have:

- Concepts Understood: Mastered the CASE statement syntax, which allows for "if-then-else" logic directly within a SQL result set.
- Skills Developed: Gained the ability to categorize data dynamically without altering the physical storage of the table.
- Practical Exposure: Learned how to handle table resets (Drop/Create) and perform mathematical operations like modulo within a query to derive meaningful business insights.